



Kueh Machine — Requirements Deck (MVP1) (Refined)

Requirement Deck — Kueh Machine (MVP1)

Slide 1 — Title

Kueh Machine

Requirement Deck (MVP1)

Slide 2 — Problem & Opportunity

- Many younger people and foreigners are unfamiliar with Singapore's traditional kueh.
 - Discoverability is low; learning is often text-heavy or scattered.
 - Opportunity: a playful, culturally grounded experience that teaches kueh through "collecting".
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Slide 3 — Product Summary

What it is: A website that simulates a gachapon machine to reveal a kueh "capsule", then teaches the user about that kueh.

Core value: Fun-first discovery + bite-sized learning (taste + history).

Primary experience loop: Insert coin → pull a lever → capsule rolls out of the machine → click on the capsule a few times to open → capsule animation → reveal kueh → learn → repeat.

Slide 4 — Goals (MVP1)

- Deliver a delightful "gachapon reveal" interaction on web.

- Teach users: **kueh name + taste + short history** for each reveal.
 - Provide a clean, accessible UI with dark mode.
 - Ensure a responsive layout across common screen sizes.
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Slide 5 — Target Audience & Personas

Target audience

- Younger audiences in Singapore
- Foreigners / newcomers interested in local culture & food

Example personas

- "New to SG Explorer": wants quick cultural context and what to try.
 - "Curious Student": enjoys gamified learning and collecting facts.
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Slide 6 — Platforms & Supported Devices

Platform

- Website (responsive)

Support targets (MVP1)

- Mobile: iOS Safari, Android Chrome (latest + 1)
 - Desktop: Chrome, Safari, Edge (latest + 1)
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Slide 7 — Scope (In)

MVP1 in-scope

- Home screen with gachapon machine UI
- "Insert coin" interaction
- Capsule roll-out + opening animation
- Kueh reveal card/page
- Kueh detail content: taste + history (short, structured)
- Light and Dark mode

- Accessibility baseline (WCAG-aligned)
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Slide 8 — Out of Scope (MVP1)

- A voting system on each kueh
 - Do you like this kueh: Yes / No / I have not tried
 - Display the stats of how many voted
 - A page with all the Kueh stats
 - “Where to buy in Singapore” listing
 - Inventory / “pokedex” collection tracking
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Slide 9 — Key User Flows

1. Primary flow

- Land on site → insert coin → pull a lever → capsule rolls out of the machine → click on the capsule a few times to open → capsule animation → reveal kueh → learn → repeat.

2. Repeat discovery

- Reveal multiple kueh consecutively without friction

3. Optional (MVP1)

- “Reveal another” CTA + simple navigation back to machine
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Slide 10 — Functional Requirements (MVP1)

FR-1: Gachapon interaction

- User can trigger a “coin insert” action.
- System prevents double-trigger during animation (state lock).

FR-2: Lever animation

- User to pull the lever

FR-3: Capsule animation sequence

- System plays roll-out animation from the machine
- User to click the capsule a few times, then display the opening animation.
- Animation must complete within an acceptable time (target < 6–8s total, configurable).

FR-4: Kueh selection logic

- System selects a kueh item from a defined dataset.
- Selection is randomized (with optional weighted rarity, if desired).
- No duplicates rule is *not required* for MVP1.

FR-5: Kueh detail screen

- Displays at minimum:
 - Kueh name
 - Short description
 - Taste profile (e.g., sweet/savory, texture)
 - Short history/cultural note
 - Image/illustration (if available)

FR-6: Navigation

- Clear CTAs:
 - "Open capsule" (if separated)
 - "Reveal another"
 - "Back to machine"

FR-7: Content dataset

- MVP1 ships with an initial curated list (recommend 10–20 kueh).
- Each entry includes structured fields (see Data Requirements).

Slide 11 — UI / UX Requirements

- Clean, minimalist interface

- The gachapon machine is the hero element
 - Clear feedback and microcopy during states:
 - Idle → inserting → pulling → rolling → reveal → details
 - Light and Dark mode support:
 - Consistent contrast ratios
 - Respects system preference (prefers-color-scheme)
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Slide 12 — Accessibility (WCAG) Requirements

MVP1 should align with WCAG 2.1 AA where feasible:

- Keyboard navigation for all interactive elements
 - Visible focus states
 - Sufficient contrast ratios (especially in dark mode)
 - Reduced motion option:
 - Respect `prefers-reduced-motion` to shorten/disable animations
 - Alt text for images; avoid text-only conveyed meaning
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Slide 13 — Performance Requirements

- First contentful load: target < 2–3s on average mobile network (best effort)
 - Animations should remain smooth (target 60fps on modern devices)
 - Optimize assets:
 - lazy-load images
 - compress images
 - bundle splitting if needed
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Slide 14 — Technical Requirements (Implementation Notes)

Frontend

- Responsive design for various screen sizes

Slide 15 — Risks & Mitigations

- **Risk:** Animations feel slow or gimmicky
 - Mitigation: keep sequence short, add skip/reduce motion
 - **Risk:** Content quality is weak / not credible
 - Mitigation: cite sources internally; keep entries reviewed
 - **Risk:** Accessibility issues with dark mode/contrast
 - Mitigation: test with contrast tooling + keyboard-only QA
 - **Risk:** Dataset too small, experience feels repetitive
 - Mitigation: ship 10–20 items; add weighted randomness later
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Slide 16 — Future Roadmap (Post-MVP)

- Places in Singapore that sell each kueh
- Inventory / collection ("kueh-dex")